

JavaScript Programming Challenges

Solve each problem using JavaScript. Focus on problem-solving, clean code, and reusable functions.

1. Student Ranking System ■■

A teacher stores exam results in an array of objects: `const students=[{name:'John',score:80},{name:'Mary',score:95},{name:'Alex',score:60},{name:'Kevin',score:88}];`

Task: Display students from highest to lowest score with ranking numbers.

Requirements: Do not modify the original array. Keep original order for equal scores.

Bonus: Display highest, lowest and average score.

2. Shopping Cart ■■

Given products with price and quantity, display each subtotal, total quantity, total price, and apply a 10% discount if total exceeds \$50.

3. Library Search ■■

A library stores books as objects (title, author, year, available). Search by title entered by the user. Display the book if found; otherwise display 'Book Not Found'. Bonus: allow partial matches.

4. Attendance Report ■■

Given an array of attendance values ('P','A','L'), count Present, Absent and Late students. Display percentages for each category.

5. Remove Duplicate Players ■■

Given an array of player names containing duplicates, create a new array containing only the first occurrence of each name while preserving order.

6. Student Dashboard ■■■

Using HTML, CSS and JavaScript, display every student object as a card. Each card shows name, age and grade with a Delete button. Bonus: add Edit.

7. Inventory Manager ■■■

Build an inventory manager using an array of product objects. Users can add, delete and update stock. Prevent negative stock. Highlight products with stock below 5.

8. Voting System ■■■

Create three buttons: Cats, Dogs and Birds. Clicking a button increases its vote count. Display live totals and the current winner. If tied, display 'Tie'.

9. Pokémon Team Builder ■■■

Allow users to build a team of up to 6 Pokémon. Prevent duplicates. Display average level, strongest Pokémon and total team power.

10. Student Management System ■■■

Create a complete application with Add, Edit, Delete and Search. Store students in an array of objects and display them in an HTML table. Bonus: sort by name or score and show summary statistics.